ICT Class 9 - Formative Assessment 1

Formative Assessment (FA-1)

- Class 9

Class: Class 9

Assessment Type: Formative Assessment (FA-1)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Python Fundamentals - Review and Advanced Concepts

Curated and developed by

Hoysala T, Computer Science Teacher

MDRS SC-32 Bahaddurghatta (Kogunde), Chitradurga Taluk & District - 577519

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General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: Written Concept Paper (20 Marks)

Section I: MCQ (5 x 1 = 5 marks)

a. What is the output of the following Python code?

```
x = [1, 2, 3, 4, 5]
print(x[1::2])
```

a. `[1, 3, 5]`

b. `[2, 4]`

c. `[1, 2, 3]`

d. Error

b. Which of the following is a valid variable name in Python?

c. `2nd_place`

d. `_score`

e. `total-marks`

f. `class`

c. What will be the output?

```
print(type(10 // 3))
```

a. `<class 'float'>`

b. `<class 'int'>`

c. `<class 'decimal'>`

d. Error

d. Which operator is used for string repetition in Python?

e. `+`

f. `*`

g. `**`

h. `%`

e. What is the output of `bool("")`?

f. `True`

g. `False`

h. `None`

i. Error

Section II: Short Answer (5 x 2 = 10 marks)

a. Differentiate between `is` and `==` operators in Python with examples.
b. Write the output of the following and explain:

```
a, b = 5, 10
a, b = b, a
print(a, b)
```

c. Explain the difference between `while` loop and `for` loop. When would you use each?
d. Write a Python program to check if a given number is a palindrome without converting it to a string.
e. What is the purpose of `break`, `continue`, and `pass` statements? Give one example of each.

Section III: Long Answer (5 x 1 = 5 marks)

a. Write a Python program that takes a list of integers and returns a new list containing only the prime numbers from the original list. Include proper function definition, docstring, and test the function with at least 3 different lists.

Part B: Hands-on Practical Lab Task (5 marks)

Write a Python program that:

- Takes a sentence as input from the user
- Counts the number of words, vowels, consonants, digits, and special characters
- Displays the frequency of each vowel
- Checks if the sentence is a palindrome (ignoring spaces and case)
- Uses functions for each counting operation

Part C: Notebook Maintenance (5 marks)

a. Neatness and organisation of notebook (2 marks)
b. Completion of all assigned classwork and homework (2 marks)
c. Proper headings, dates, and indexing (1 mark)

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
Pattern Recognition	Cannot identify similarities or trends in data or problems	Identifies patterns with prompts or scaffolding	Independently identifies and applies patterns to solve new problems

Abstraction	Includes all details, cannot filter irrelevant information	Removes some irrelevant details with guidance	Filters to essential information, focuses on what matters
Algorithm Design	Provides random or disordered steps	Creates sequential but incomplete step-by-step solutions	Designs clear, ordered, complete algorithms with start and end
Debugging	Gives up when errors occur	Finds and fixes errors with help from teacher or peers	Systematically tests, identifies, and fixes errors independently
Evaluation	Does not check if solution works	Tests solution with guidance	Evaluates efficiency and correctness of own and others' solutions

Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: "What I learned" and "What was hard" (2 marks)
- Real-life connections: "Where I see this outside school" (1 mark)

ICT Class 9 - Formative Assessment 2

Formative Assessment (FA-2)

- Class 9

Class: Class 9

Assessment Type: Formative Assessment (FA-2)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Data Structures - Lists, Tuples, Dictionaries, Sets

Curated and developed by

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Part A: Written Concept Paper (20 Marks)

Section I: MCQ (5 x 1 = 5 marks)

a. What is the output?

```
s = {1, 2, 3, 2, 1}
print(len(s))
```

a. 5

b. 3

c. 2

d. Error

b. Which method returns a list of tuples representing key-value pairs in a dictionary?

c. `keys()`

d. `values()`

e. `items()`

f. `pairs()`

c. What will be the output?

```
lst = [[1, 2], [3, 4], [5, 6]]
result = [x for row in lst for x in row if x % 2 == 0]
print(result)
```

a. `[1, 3, 5]`

b. `[2, 4, 6]`

c. `[2, 4]`

d. `[[2], [4], [6]]`

d. Which of the following is true about tuples?

e. Tuples are mutable

f. Tuples can contain mixed data types

g. Tuples are faster than lists

h. Both B and C

e. What does `dict.fromkeys(['a', 'b', 'c'], 0)` return?

f. `{'a': 0, 'b': 0, 'c': 0}`

g. `{'a': 'b': 'c': 0}`

h. `['a', 'b', 'c']`
i. Error

Section II: Short Answer (5 x 2 = 10 marks)

a. Differentiate between a **list** and a **tuple**. Give two real-world use cases where a tuple is preferred over a list.
b. Write the output of:

```
d1 = {"x": 1, "y": 2}
d2 = {"y": 3, "z": 4}
d1.update(d2)
print(d1)
```

c. Explain the difference between `remove()`, `pop()`, and `del` for lists with examples.
d. Write a Python program to find the intersection and union of two sets without using built-in set operations.
e. What is **list comprehension**? Write a list comprehension to generate all Pythagorean triplets `(a, b, c)` where `a < b < c` and `a² + b² = c²` for values up to 20.

Section III: Long Answer (5 x 1 = 5 marks)

a. Write a Python program that implements a **word frequency counter** using dictionaries. The program should:

- Read a paragraph of text
- Count the frequency of each word (case-insensitive, ignoring punctuation)
- Sort words by frequency in descending order
- Display the top 5 most frequent words with their counts
- Find and display words that appear exactly once

Part B: Hands-on Practical Lab Task (5 marks)

Write a Python program that:

- Creates a list of 15 student records as dictionaries with keys: name, roll_no, marks (out of 100)
- Implements functions to:
  - Find the topper (highest marks)
  - Calculate the class average
  - Find students who scored below average
  - Sort students by marks in descending order
  - Display results in a formatted table

Part C: Notebook Maintenance (5 marks)

a. Neatness and organisation of notebook (2 marks)
b. Completion of all assigned classwork and homework (2 marks)
c. Proper headings, dates, and indexing (1 mark)

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
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Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: "What I learned" and "What was hard" (2 marks)
- Real-life connections: "Where I see this outside school" (1 mark)

ICT Class 9 - Formative Assessment 3

Formative Assessment (FA-3)

- Class 9

Class: Class 9

Assessment Type: Formative Assessment (FA-3)

Total Marks: 30

Duration: 90 minutes

Topics Covered: SQL - Database Management

Curated and developed by

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General Instructions

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Part A: Written Concept Paper (20 Marks)

Section I: MCQ (5 x 1 = 5 marks)

a. Which SQL clause is used to combine rows from two or more tables?
b. `GROUP BY`
c. `JOIN`
d. `UNION`
e. `COMBINE`
b. What is the result of `SELECT COUNT(DISTINCT department) FROM Employees;`?
c. Counts all rows
d. Counts only distinct departments
e. Counts only NULL values
f. Error
c. Which of the following is a DDL command?
d. `SELECT`
e. `INSERT`
f. `ALTER`
g. `UPDATE`
d. What does the `LIKE` operator do in SQL?
e. Exact match
f. Pattern matching
g. Range comparison
h. NULL check
e. Which aggregate function ignores NULL values?
f. `COUNT(*)`
g. `COUNT(column_name)`
h. `SUM(column_name)`
i. Both B and C

Section II: Short Answer (5 x 2 = 10 marks)

a. Write a SQL query to create a table `Library` with columns: `book_id` (INT PRIMARY KEY AUTO_INCREMENT), `title` (VARCHAR 100), `author` (VARCHAR 50), `genre` (VARCHAR 30), `price` (DECIMAL), `available` (BOOLEAN).
b. Differentiate between `WHERE` and `HAVING` clauses with two examples each.
c. Write SQL queries for a table `Students(roll, name, class, section, marks)`:
d. Find students whose name starts with 'A'
e. Find students with marks between 50 and 80
f. Find students in sections 'A' or 'B'
g. Count students in each class
d. What is a **Foreign Key**? Explain its role in maintaining referential integrity with an example.
e. Write a SQL query to display the second highest salary from an `Employees` table without using `LIMIT` or `TOP`.

Section III: Long Answer (5 x 1 = 5 marks)

a. Given the database schema:

```
Students(roll_no, name, class, section)
Subjects(sub_id, sub_name, max_marks)
Marks(roll_no, sub_id, marks_obtained)
```

Write SQL queries to:

a. Find the percentage of each student (total marks obtained / total max marks — 100)
b. Find the subject with the highest average score
c. List students who scored above 90% in any subject
d. Find the class topper (highest overall percentage)
e. Display a rank list of all students by total marks

Part B: Hands-on Practical Lab Task (5 marks)

Write a Python program that:

- Connects to an SQLite database
- Creates tables `School(roll, name, class, section)` and `Results(roll, sub1, sub2, sub3, sub4, sub5)`
- Inserts at least 8 student records with their marks
- Performs and displays:
  - Average marks per student
  - Subject with highest class average
  - Students who need improvement (average below 60%)
  - Grade assignment (A: 90+, B: 75-89, C: 60-74, D: 45-59, F: below 45)

Part C: Notebook Maintenance (5 marks)

a. Neatness and organisation of notebook (2 marks)
b. Completion of all assigned classwork and homework (2 marks)

c. Proper headings, dates, and indexing (1 mark)

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

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Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

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- Screenshots or photos of work completed in class (2 marks)
- Written reflections: "What I learned" and "What was hard" (2 marks)
- Real-life connections: "Where I see this outside school" (1 mark)

ICT Class 9 - Formative Assessment 4

Formative Assessment (FA-4)

- Class 9

Class: Class 9

Assessment Type: Formative Assessment (FA-4)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Web Technologies - HTML, CSS, JavaScript

Curated and developed by

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General Instructions

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4. Marks are indicated against each question.

Part A: Written Concept Paper (20 Marks)

Section I: MCQ (5 x 1 = 5 marks)

a. Which HTML5 tag is used to define a navigation section?
b. <menu>
c. <nav>
d. <navigation>
e. <links>
b. Which CSS property is used to create a flexible layout?
c. `display: block`
d. `display: inline`
e. `display: flex`
f. `display: table`
c. What does `document.getElementById("demo").innerHTML = "Hello";` do?
d. Changes the CSS of element
e. Changes the HTML content of element
f. Deletes the element
g. Creates a new element
d. Which event occurs when the user clicks on an HTML element?
e. `onmouseover`
f. `onchange`
g. `onclick`
h. `onmouseclick`
e. Which CSS unit is relative to the font-size of the root element?
f. `em`
g. `rem`
h. `px`
i. `%`

Section II: Short Answer (5 x 2 = 10 marks)

a. Differentiate between **inline CSS**, **internal CSS**, and **external CSS**. When would you use each?
b. Write the HTML and CSS code to create a responsive navigation bar using flexbox that collapses into a hamburger menu on mobile screens (write the structure only).
c. Explain the CSS **box model** with a labelled diagram. How does `box-sizing: border-box` change the default behaviour?
d. Write JavaScript code to:
e. Validate that a form field is not empty
f. Display an alert with the field value if valid
g. Show an error message if empty
e. What is the difference between `var`, `let`, and `const` in JavaScript? Give one example of each.

Section III: Long Answer (5 x 1 = 5 marks)

a. Create a complete web page for a **Student Management System** that includes:
b. An HTML form with fields: Name, Class, Section, Roll Number, and three subject marks
c. CSS styling with a professional colour scheme, styled form inputs, and a submit button
d. JavaScript to:
   - Calculate the total and percentage when the user clicks a "Calculate" button
   - Assign a grade based on the percentage
   - Display the results in a styled card below the form
   - Validate all fields before submission

Part B: Hands-on Practical Lab Task (5 marks)

Create a **Quiz Application** web page that:

- Displays one question at a time with 4 multiple-choice options
- Uses JavaScript to track the current question and score
- Highlights correct/incorrect answers with colour changes
- Shows a progress bar indicating current question out of total
- Displays the final score with a message at the end
- Uses CSS for attractive styling with animations

Part C: Notebook Maintenance (5 marks)

a. Neatness and organisation of notebook (2 marks)
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Part D: Computational Thinking Skills (Internal Assessment)

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ICT Class 9 - Practical Lab Exam 1

Practical Lab Exam 1

- Class 9

Class: Class 9

Assessment Type: Practical Lab Exam 1

Total Marks: 20

Duration: 45 minutes

Topics Covered: Python Programming and Data Structures

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Lab Tasks (15 marks)

Task 1: Student Grade Manager (5 marks)

Write a Python program that:

- Stores student data as a list of dictionaries with keys: name, roll_no, subjects (dict of sub: marks)
- Implements functions for:
  - Adding a new student with marks in 5 subjects
  - Updating marks for an existing student
  - Calculating grade based on average (A: 90+, B: 75-89, C: 60-74, D: 45-59, F: below 45)
  - Finding the class topper and subject toppers
  - Displaying a formatted report card for all students
- Uses proper input validation and error handling

Task 2: Inventory Management System (5 marks)

Write a Python program that:

- Uses a dictionary to store inventory items with keys: item_id, name, quantity, price, category
- Implements a menu-driven interface with options:
  - Add new item
  - Update stock (increase/decrease)
  - Search items by name or category
  - Calculate total inventory value
  - Display items sorted by price or quantity
  - Generate a low-stock alert (quantity below 5)
- Includes a function to export inventory data to a text file

Task 3: Pattern Generator (5 marks)

Write a Python program that generates the following patterns using nested loops:

- Pattern A: Right-angled triangle of stars (5 rows)

```
*
**
***
****
*****
```

- Pattern B: Pyramid pattern (5 rows)

```
    *
   ***
  *****
 *******
*********
```

- Pattern C: Number pattern (5 rows)

```
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```

- The program should use functions for each pattern and allow the user to choose which pattern to display
- Allow custom input for the number of rows (1-10)

Viva Voce (5 marks)

1. What is the difference between a **function definition** and a **function call**? Give an example of each. (1 mark)
2. Explain the difference between **mutable** and **immutable** data types. Give two examples of each. (1 mark)
3. How does Python handle memory when you create a **shallow copy** vs a **deep copy** of a nested list? (1 mark)
4. What is the purpose of the `__init__` method in a Python class? Can you create an object without it? (1 mark)
5. Explain the difference between **linear search** and **binary search**. When would you choose one over the other? (1 mark)

Part D: Computational Thinking Skills (Internal Assessment)

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Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

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- Real-life connections: "Where I see this outside school" (1 mark)

ICT Class 9 - Practical Lab Exam 2

Practical Lab Exam 2

- Class 9

Class: Class 9

Assessment Type: Practical Lab Exam 2

Total Marks: 20

Duration: 45 minutes

Topics Covered: SQL with Python and Web Development

Curated and developed by

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Lab Tasks (15 marks)

Task 1: SQL Database Operations with Python (5 marks)

Write a Python program using SQLite that:

- Creates a database `school.db` with tables:
  - `Students(roll_no INTEGER PRIMARY KEY, name TEXT, class TEXT, section TEXT, admission_date DATE)`
  - `Subjects(sub_id INTEGER PRIMARY KEY, sub_name TEXT, max_marks INTEGER)`
  - `Results(roll_no INTEGER, sub_id INTEGER, marks_obtained INTEGER, FOREIGN KEY references Students and Subjects)`
- Inserts at least 6 students and 4 subjects with result data
- Executes and displays:
  - Students with marks above 80 in any subject
  - Subject-wise average and highest marks
  - Students ranked by total marks
  - Class-wise performance summary (total students, average marks per class)
- Updates one student's marks and displays the before/after comparison

Task 2: HTML and CSS Web Page (5 marks)

Create a complete HTML page `school_portal.html` with an external CSS file `portal.css` that includes:

- A styled header with school name and navigation links (Home, About, Academics, Contact)
- A hero section with a welcome message and call-to-action button
- A table displaying 6 students with columns: Name, Class, Section, Marks, Grade
- The table should have:
  - Alternating row colours (zebra striping)
  - Hover effect on rows
  - Proper borders, spacing, and responsive design
- A contact form with fields: Name, Email, Subject, Message, and a Submit button
- A styled footer with copyright information
- CSS should use Flexbox or Grid for layout

Task 3: Interactive Web Application (5 marks)

Create a web page `todo_app.html` with embedded JavaScript that:

- Allows users to add new tasks with a title and priority (High, Medium, Low)
- Displays tasks in a list with colour-coded priority indicators
- Allows marking tasks as complete (strikethrough effect)
- Allows deleting tasks
- Shows a summary: total tasks, completed tasks, pending tasks
- Uses CSS for attractive styling with smooth transitions/animations
- Stores tasks in an array (in-memory) and re-renders the list on any change

Viva Voce (5 marks)

1. What is the difference between `INNER JOIN` and `LEFT JOIN`? Give one example of each. (1 mark)
2. How does CSS specificity work? Which rule takes priority: inline, internal, or external CSS? Explain with an example. (1 mark)
3. What is the purpose of a **Foreign Key** in SQL? How does it maintain data integrity between tables? (1 mark)
4. Explain the difference between `WHERE` and `HAVING` in SQL. When would you use each? (1 mark)
5. What is the difference between **responsive design** and **adaptive design** in web development? Give one example of each approach. (1 mark)

Part D: Computational Thinking Skills (Internal Assessment)

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ICT Class 9 - Summative Assessment 1

Summative Assessment (SA-1)

- Class 9

Class: Class 9

Assessment Type: Summative Assessment (SA-1)

Total Marks: 40

Duration: 2 hours

Topics Covered: Python Fundamentals, Data Structures, SQL, Web Technologies

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General Instructions

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3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: MCQ (10 x 1 = 10 marks)

a. What is the output?

```
x = [1, 2, 3, 4, 5]
print(x[::2] + x[1::2])
```

a. `[1, 3, 5, 2, 4]`

b. `[1, 2, 3, 4, 5]`

c. `[1, 3, 5, 1, 3, 5]`

d. Error

b. Which of the following is the correct way to create a dictionary?

c. `dict = {"a": 1, "b": 2}`

d. `dict = [("a", 1), ("b", 2)]`

e. `dict = {["a"]: 1}`

f. `dict = dict("a"=1)`

c. What will be the output?

```
s = "Python"
print(s[-3:])
```

a. `"tho"`

b. `"hon"`

c. `"tho"`

d. `"no"`

d. Which SQL command removes a table completely from the database?

e. `DELETE TABLE`

f. `REMOVE TABLE`

g. `DROP TABLE`

h. `TRUNCATE TABLE`

e. Which HTML tag is used to define an unordered list?

f. `<ol>`

g. `<ul>`

h. `<li>`

i. `<list>`

f. What is the time complexity of searching an element in a sorted list using binary search?
g. O(n)
h. O(log n)
i. $O(n^2)$
j. O(1)
g. Which CSS property is used to change the font family?
h. `text-font`
i. `font-family`
j. `font-style`
k. `text-family`
h. What will be the output?

```
d = {"x": 1, "y": 2, "z": 3}
print(list(d.values()))
```

a. `["x", "y", "z"]`
b. `[1, 2, 3]`
c. `("x", "y", "z")`
d. Error
i. Which JavaScript method selects an element by its ID?
j. `document.querySelector()`
k. `document.getElementById()`
l. `document.getElement()`
m. `document.findById()`
j. What does `COUNT(*)` do in SQL?
k. Counts only non-NULL values
l. Counts all rows including NULL values
m. Counts only distinct values
n. Error

Part B: Short Answer (8 x 2 = 16 marks)

a. Write a Python function `merge_dicts(d1, d2)` that merges two dictionaries. If a key exists in both, the values should be added together. Test with `{"a": 1, "b": 2}` and `{"b": 3, "c": 4}`.
b. Write the output of:

```
lst = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]
result = [lst[i][i] for i in range(len(lst))]
print(result)
```

Explain the logic.
c. Write SQL queries for a table `Employees(emp_id, name, department, salary, joining_date)`:
d. Find employees who joined in the last 2 years
e. Find the second highest salary

f. Find departments where average salary exceeds 50000

d. Differentiate between `DELETE`, `TRUNCATE`, and `DROP` with one example each.

e. Explain the CSS **Flexbox** layout model. Write CSS to create a three-column layout where the middle column is twice the width of the side columns.

f. What is **responsive web design**? Explain three techniques used to make a web page responsive.

g. Write a Python program to implement a **stack** using a list with operations: push, pop, peek, is_empty, and display.

h. Write a SQL query to find the **nth highest salary** from an `Employees` table. Write it for `n = 5`.

Part C: Long Answer (4 x 3 = 12 marks)

a. Write a complete Python program that implements a **Library Management System** using dictionaries and lists. The program should:

- Store book details: title, author, ISBN, available copies
- Allow operations: add book, issue book, return book, search by title/author
- Display all issued books and their due dates
- Calculate and display fines for overdue books (₹5 per day)
- Use functions for each operation with proper error handling

b. Write SQL queries for an **E-Commerce Database** with tables:

```
Customers(cust_id, name, email, city)
Products(prod_id, name, price, category, stock)
Orders(order_id, cust_id, order_date, total_amount)
Order_Items(order_id, prod_id, quantity, price)
```

Write queries to:

a. Find the top 5 customers by total spending

b. Find products that have never been ordered

c. Calculate monthly revenue for the current year

d. Find the most popular product category

e. List customers who ordered more than 3 different products

c. Create a complete web page for a **Student Report Card Generator** that:

- Has a form to enter: Student Name, Class, Section, Roll Number, and marks for 5 subjects
- Uses CSS Grid layout for the form
- JavaScript to:
  - Validate all inputs (marks must be between 0 and 100)
  - Calculate total, percentage, and grade
  - Generate a styled report card below the form
  - Include a print button that opens the print dialog

d. Write a Python program that implements a **Contact Book** using:

- File handling to persist data between runs
- Dictionary of lists to store contacts

- Operations: add, search, update, delete, display all contacts
- Search by name (partial match), phone number, or email
- Sort contacts alphabetically or by creation date
- Export contacts to a CSV file

Part D: Application Based (2 x 1 = 2 marks)

a. A school wants to create a system that tracks student attendance. Explain how:

- Python data structures (dictionaries, lists) can store daily attendance
- SQL database can store historical attendance data with proper schema
- A web interface (HTML/CSS/JS) can display attendance reports
- Include the key components needed for each technology

b. An online bookstore needs an inventory system. Analyse how:

- SQL can maintain the book catalog with proper relationships
- Python can process bulk imports from CSV files
- A web form can allow adding new books
- Consider at least one SQL query that would be commonly used

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
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Debugging	Gives up when errors occur	Finds and fixes errors with help from teacher or peers	Systematically tests, identifies, and fixes errors independently
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Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: “What I learned” and “What was hard” (2 marks)
- Real-life connections: “Where I see this outside school” (1 mark)

ICT Class 9 - Summative Assessment 2

Summative Assessment (SA-2)

- Class 9

Class: Class 9

Assessment Type: Summative Assessment (SA-2)

Total Marks: 40

Duration: 2 hours

Topics Covered: Algorithms, Cyber Security, Emerging Technologies, Computational Thinking

Curated and developed by

Hoysala T, Computer Science Teacher

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General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: MCQ (10 x 1 = 10 marks)

a. What is the time complexity of bubble sort in the worst case?
b. O(n)
c. O(n log n)
d. $O(n^2)$
e. O(log n)
b. Which of the following is NOT a characteristic of a good algorithm?
c. Finiteness
d. Definiteness
e. Ambiguity
f. Effectiveness
c. What does phishing refer to?
d. A type of malware
e. A social engineering attack to steal credentials
f. A hardware malfunction
g. A network protocol
d. Which technology is the foundation of Bitcoin?
e. Cloud Computing
f. Artificial Intelligence
g. Blockchain
h. Big Data
e. What is the primary purpose of a firewall?
f. To speed up internet
g. To filter network traffic and block threats
h. To store data
i. To compile programs
f. In a flowchart, which shape represents an input/output operation?
g. Rectangle
h. Diamond

i. Parallelogram
j. Oval
g. Which of the following is an example of **phishing**?
h. A virus infecting a computer
i. An email asking for password pretending to be from a bank
j. A hardware crash
k. A software update
h. What is **Machine Learning**?
i. A type of hardware
j. A subset of AI that enables systems to learn from data
k. A programming language
l. A database system
i. Which cryptographic technique uses the same key for encryption and decryption?
j. Asymmetric encryption
k. Symmetric encryption
l. Hashing
m. Digital signature
j. What is the full form of IoT?
k. Internet of Technologies
l. Internet of Things
m. Integration of Things
n. Interface of Technology

Part B: Short Answer (8 x 2 = 16 marks)

a. Write an algorithm (in pseudocode) to find the second smallest element in an unsorted list. Analyse its time complexity.
b. Differentiate between **encryption** and **decryption**. Give one real-world example of each.
c. Explain three types of **malware** with their characteristics and one prevention method each.
d. What is **Big Data**? Explain the 5 V's of Big Data with one sentence each.
e. Write a Python program to implement **selection sort** and trace it on `[64, 25, 12, 22, 11]`.
f. Explain **two-factor authentication**. Why is it more secure than a single password?
g. What is **cloud computing**? Differentiate between **public cloud**, **private cloud**, and **hybrid cloud** with one example each.
h. Write the pseudocode for a **binary search** algorithm. Trace it on `[2, 5, 8, 12, 16, 23, 38, 56, 72, 91]` to find `23`.

Part C: Long Answer (4 x 3 = 12 marks)

a. Write a complete Python program that implements **Merge Sort**. Trace the algorithm on `[38, 27, 43, 3, 9, 82, 10]` showing each divide and merge step. Analyse the time complexity for best, average, and worst cases.

b. A school wants to implement a **cyber safety policy**. Write:

- Five common cyber threats students may face
- Detailed prevention measures for each threat
- A step-by-step incident response plan if a student's account is compromised
- How the school can use technology to monitor and prevent cyberbullying

c. Explain the following emerging technologies with detailed examples:

- **Artificial Intelligence**: How it works, two applications in education, and ethical concerns
- **Internet of Things (IoT)**: Components, how sensors work, and one smart home application
- **Blockchain**: Structure, how transactions are verified, and one application beyond cryptocurrency

d. Design a **Sorting Algorithm Visualiser** project concept. Write:

- The problem statement and objectives
- A Python program structure using dictionaries and lists to simulate sorting steps
- How bubble sort, insertion sort, and selection sort would be implemented
- How a web interface could display the sorting process with JavaScript animations

Part D: Application Based (2 x 1 = 2 marks)

a. A social media company wants to protect user data. Explain how:

- Encryption can secure messages in transit
- Firewalls and intrusion detection systems protect servers
- Strong password policies and 2FA protect individual accounts
- What regulations must the company follow regarding data privacy

b. A healthcare facility wants to digitise patient records. Analyse how:

- Cloud computing can provide scalable storage for medical data
- IoT devices can monitor patients remotely and send data
- Cyber security measures are critical for protecting sensitive health information
- What ethical considerations arise from collecting patient data

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)

Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
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Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

- Screenshots or photos of work completed in class (2 marks)
- Written reflections: "What I learned" and "What was hard" (2 marks)
- Real-life connections: "Where I see this outside school" (1 mark)

ICT Class 9 - Unit Test 1

Unit Test 1

- Class 9

Class: Class 9

Assessment Type: Unit Test 1

Total Marks: 10

Duration: 30 minutes

Topics Covered: Python Fundamentals

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Hoysala T, Computer Science Teacher

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General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Section I: MCQ (5 x 1 = 5 marks)

a. What is the output of `print(2 ** 3 ** 2)`?
b. 64
c. 512
d. 36
e. Error
b. Which of the following is NOT a valid Python data type?
c. `list`
d. `tuple`
e. `array`
f. `dict`
c. What will be the output?

```
x = "Hello"
y = x[:]
print(y[1] + y[3])
```

a. `el`
b. `el`
c. `ell`
d. Error
d. Which keyword is used to handle exceptions in Python?
e. `catch`
f. `except`
g. `handle`
h. `error`
e. What is the output of `print(not True or False and True)`?
f. `True`
g. `False`
h. `None`
i. Error

Section II: Short Answer (5 x 1 = 5 marks)

a. Write the output of:

```
for i in range(1, 10, 3):
    print(i, end=" ")
```

b. Write a Python function `count_vowels(s)` that returns the number of vowels in string `s`. Call it with `"Hello World"` and show the output.

c. What is the difference between `range(5)` and `range(1, 5)`? Write the output of each.

d. Write a program to check if a number is an Armstrong number ($153 = 1^3 + 5^3 + 3^3$).

e. Explain what happens when you divide by zero in Python. How can you handle it?

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
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Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

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- Written reflections: “What I learned” and “What was hard” (2 marks)
- Real-life connections: “Where I see this outside school” (1 mark)

ICT Class 9 - Unit Test 2

Unit Test 2

- Class 9

Class: Class 9

Assessment Type: Unit Test 2

Total Marks: 10

Duration: 30 minutes

Topics Covered: Data Structures

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General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
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Section I: MCQ (5 x 1 = 5 marks)

a. What will be the output?

```
d = {"a": [1, 2], "b": [3, 4]}
d["a"].append(5)
print(d["a"])
```

a. `[1, 2]`

b. `[1, 2, 5]`

c. `[5, 1, 2]`

d. Error

b. Which method is used to sort a list in place?

c. `sorted()`

d. `sort()`

e. `order()`

f. `arrange()`

c. What is the output?

```
t = ([1, 2], [3, 4])
t[0].append(5)
print(t)
```

a. `([1, 2], [3, 4])`

b. `([1, 2, 5], [3, 4])`

c. Error (tuples are immutable)

d. `([5, 1, 2], [3, 4])`

d. Which of the following creates a shallow copy of a list?

e. `copy = list`

f. `copy = list[:]`

g. `copy = list.copy()`

h. Both B and C

e. What does `set([1, 2, 2, 3, 3, 3]).pop()` return?

f. An arbitrary element from the set

g. The last element added

h. Always 1

i. Error

Section II: Short Answer (5 x 1 = 5 marks)

a. Write the output of:

```
nums = [3, 1, 4, 1, 5, 9, 2, 6]
print(sorted(nums, reverse=True)[:3])
```

b. Write a program to merge two dictionaries. If a key exists in both, sum the values.

c. What will be the output and why?

```
lst = [1, 2, 3]
lst2 = lst
lst2.append(4)
print(lst)
```

d. Differentiate between `pop()` and `remove()` for dictionaries with examples.

e. Write a program to flatten a nested list `[[1, 2], [3, 4], [5, 6, 7]]` into a single list.

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
Decomposition	Needs help breaking down complex tasks into smaller parts	Can break down tasks with some guidance or prompts	Independently breaks down complex problems into manageable sub-tasks
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Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

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- Written reflections: “What I learned” and “What was hard” (2 marks)
- Real-life connections: “Where I see this outside school” (1 mark)

ICT Class 9 - Unit Test 3

Unit Test 3

- Class 9

Class: Class 9

Assessment Type: Unit Test 3

Total Marks: 10

Duration: 30 minutes

Topics Covered: SQL

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General Instructions

1. All questions are compulsory.
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4. Marks are indicated against each question.

Section I: MCQ (5 x 1 = 5 marks)

a. Which SQL command is used to add a new column to an existing table?
b. `ADD COLUMN`
c. `ALTER TABLE ... ADD`
d. `INSERT INTO`
e. `UPDATE TABLE`
b. What is the result of `SELECT * FROM Students ORDER BY marks DESC LIMIT 3;`?
c. Top 3 students by marks
d. Bottom 3 students by marks
e. All students sorted by marks
f. Error
c. Which of the following is NOT an aggregate function in SQL?
d. `SUM()`
e. `COUNT()`
f. `ORDER()`
g. `AVG()`
d. What does `IS NULL` check for in SQL?
e. Empty string
f. Zero value
g. Missing/unknown value
h. Undefined variable
e. Which SQL clause is used to group rows sharing common values?
f. `WHERE`
g. `GROUP BY`
h. `ORDER BY`
i. `HAVING`

Section II: Short Answer (5 x 1 = 5 marks)

a. Write SQL queries for a table `Products(pid, name, price, category, stock)`:

b. Find products with stock less than 10
c. Count products in each category
d. Find the average price per category

b. Differentiate between `INNER JOIN` and `LEFT JOIN` with an example.
c. Write a query to display the third highest salary from `Employees` table.
d. What is a **Primary Key**? Can a table have multiple primary keys? Explain.
e. Write SQL to create a table and insert data, then display only duplicate records from the table.

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
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Student Name: ***Date:***

Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

Students maintain a Computing Journal throughout the term. Award marks based on:

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- Real-life connections: "Where I see this outside school" (1 mark)

ICT Class 9 - Unit Test 4

Unit Test 4

- Class 9

Class: Class 9

Assessment Type: Unit Test 4

Total Marks: 10

Duration: 30 minutes

Topics Covered: Web Technologies - HTML, CSS, JavaScript

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General Instructions

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Section I: MCQ (5 x 1 = 5 marks)

a. Which HTML5 element is used to embed a video?
b. `<media>`
c. `<video>`
d. `<embed>`
e. `<movie>`
b. Which CSS property changes the order of flex items?
c. `flex-direction`
d. `order`
e. `flex-order`
f. `position`
c. What is the output of `typeof null` in JavaScript?
d. `"null"`
e. `"object"`
f. `"undefined"`
g. Error
d. Which method adds an element to the end of an array in JavaScript?
e. `push()`
f. `append()`
g. `add()`
h. `insert()`
e. Which CSS property creates a shadow effect on text?
f. `text-shadow`
g. `shadow-text`
h. `font-shadow`
i. `text-effect`

Section II: Short Answer (5 x 1 = 5 marks)

a. Write HTML code to create a form with email validation using the `pattern` attribute.

b. What is the difference between `display: none` and `visibility: hidden` in CSS?
c. Write JavaScript to find the largest element in an array `[23, 56, 12, 89, 45]`.
d. Explain the purpose of the `meta viewport` tag for responsive design.
e. Write CSS media queries to change the background colour to light blue on screens smaller than 600px.

Part D: Computational Thinking Skills (Internal Assessment)

Instructions: Observe the student during the practical task and rate each computational thinking skill on a 3-point scale.

CT Skill	Emerging (1)	Developing (2)	Proficient (3)
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Student Name: ***Date:***
Total CT Score: */ 18* ***Teacher Remarks:***

Part E: Computing Journal (Portfolio)

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